

Chapter Summary

This chapter places emphasis on determining the airworthiness of the airplane, preflight visual inspection, managing risk and pilot-available resources, safe surface-based operations, and the adherence to and proper use of the AFM/POH and checklists. The pilot should ensure that the airplane is in a safe condition for flight, and it meets all the regulatory requirements of 14 CFR part 91. A pilot also needs to recognize that flight safety includes proper flight preparation and having the experience to manage the risks associated with the expected conditions. An effective and continuous assessment and mitigation of the risks and appropriate utilization of resources goes a long way provided the pilot honestly evaluates their ability to act as PIC.

Airplane Flying Handbook (FAA-H-8083-3C)

Chapter 3: Basic Flight Maneuvers

Introduction

Airplanes operate in an environment that is unlike an automobile. Drivers tend to drive with a fairly narrow field of view and focus primarily on forward motion. Beginning pilots tend to practice the same. Flight instructors face the challenge of teaching beginning pilots about attitude awareness; which requires understanding the motions of flight. An airplane rotates in bank, pitch, and yaw while also moving horizontally, vertically, and laterally. The four fundamentals (straight-and-level flight, turns, climbs, and descents) are the principal maneuvers that control the airplane through the six motions of flight.

The Four Fundamentals

To master any subject, one should first master the fundamentals. For flying, this includes straight-and-level flight, turns, climbs, and descents. All flying tasks are based on these maneuvers, and an attempt to move on to advanced maneuvers prior to mastering the four fundamentals hinders the learning process.

Consider the following: a takeoff is a combination of a ground roll, which may transition to a brief period of straight-and-level flight, and a climb. After-departure includes the climb and turns toward the first navigation fix and is followed by straight-and-level flight. The preparation for landing at the destination may include combinations of descents, turns, and straight-and-level flight. In a typical general aviation (GA) airplane, the final approach ends with a transition from descent to straight-and-level while slowing for the touchdown and ground roll.

The flight instructor needs to impart competent knowledge of these basic flight maneuvers so that the beginning pilot is able to combine them at a performance level that at least meets the Federal Aviation Administration (FAA) Airman Certification Standards (ACS) or Practical Test Standards (PTS). As the beginning pilot progresses to more complex flight maneuvers, any deficiencies in the mastery of the four fundamentals are likely to become barriers to effective and efficient learning.

Effect and Use of Flight Controls

The airplane flies in an environment that allows it to travel up and down as well as left and right. Note that movement *up or down* depends on the flight conditions. If the airplane is right-side up relative to the horizon, forward control stick or wheel (elevator control) movement will result in a loss of altitude. If the same airplane is upside-down relative to the horizon that same forward control movement will result in a gain of altitude. [Figure 3-1] The following discussion considers the pilot's frame of reference with respect to the flight controls. [Figure 3-2]

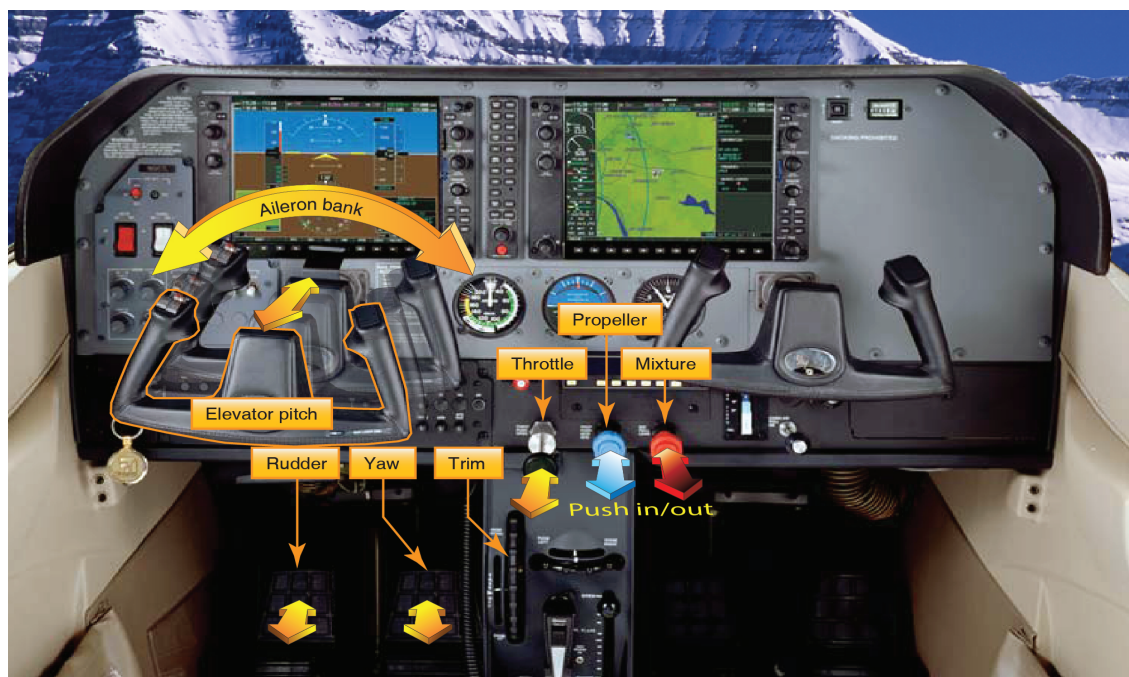


Figure 3-1. Basic flight controls and instrument panel.

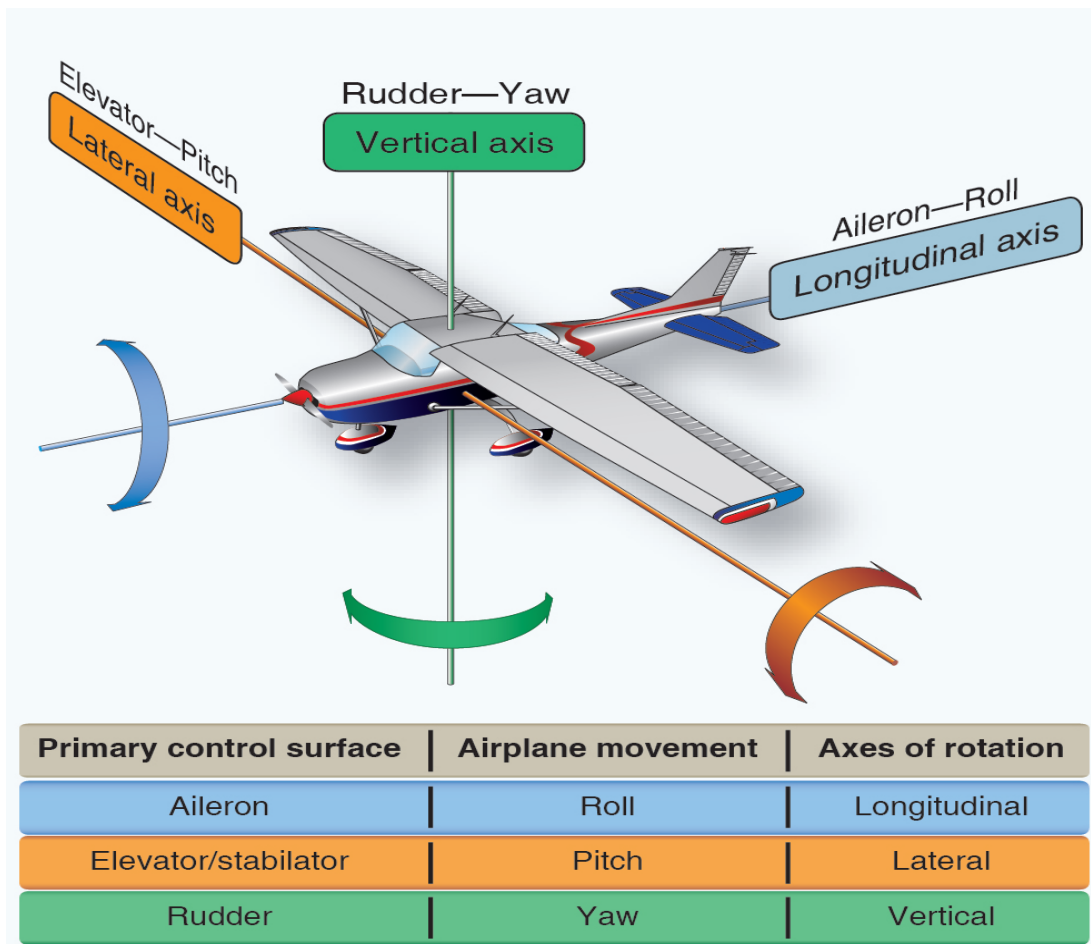


Figure 3-2. *The pilot is always considered the referenced center of effect as the flight controls are used.*

With the pilot's hand:

- When pulling the elevator pitch control toward the pilot, which is an aft movement of the control wheel, yoke, control stick, or side stick controller (referred to as adding back pressure), the airplane's nose will rotate backwards relative to the pilot around the pitch (lateral) axis of the airplane. Think of this movement from the pilot's feet to the pilot's head.
- When pushing elevator pitch control toward the instrument panel, (referred to as increasing forward pressure), the airplane rotates the nose forward relative to the pilot around the pitch axis of the airplane. Think of this movement from the pilot's head to the pilot's feet.
- When right pressure is applied to the aileron control, which rotates the control wheel or yoke clockwise, or deflects the control stick or side stick to the right, the airplane's right wing banks (rolls) lower in relation to the pilot. Think of this movement from the pilot's head to the pilot's right hip.
- When left pressure is applied to the aileron control, which rotates the control wheel or yoke counterclockwise, or deflects the control stick or side stick to the left, the airplane's left wing banks (rolls) lower in relation to the pilot. Think of this movement from the pilot's head to the pilot's left hip.

With the pilot's feet:

- When forward pressure is applied to the right rudder pedal, the airplane's nose moves (yaws) to the right in relation to the pilot. Think of this movement from the pilot's left shoulder to the pilot's right shoulder.
- When forward pressure is applied to the left rudder pedal, the airplane's nose moves (yaws) to the left in relation to the pilot. Think of this movement from the pilot's right shoulder to the pilot's left shoulder.

While in flight, the control surfaces remain in a fixed position as long as all forces acting upon them remain balanced. Resistance to movement increases as airspeed increases and decreases as airspeed decreases. Resistance also increases as the controls move away from a streamlined position. While maneuvering the airplane, it is not the amount of control surface displacement the pilot needs to consider, but rather the application of flight control pressures that give the desired result.

The pilot should hold the pitch and roll flight controls (aileron and elevator controls, yoke, stick, or side-stick control) lightly with the fingers and not grab or squeeze them with the entire hand. When flight control pressure is applied to change a control surface position, the pilot should exert pressure on the aileron and elevator controls with the fingers only. This is an important concept and habit to learn. A common error with beginning pilots is that they grab the aileron and elevator controls with a closed palm with such force that sensitive feeling is lost. Pilots may wish to consider this error at the onset of training as it prevents the development of “feel,” which is an important aspect of airplane control.

So that slight rudder pressure changes can be felt, both heels should support the weight of the pilot's feet on the floor with the ball of each foot touching the individual rudder pedals. The legs and feet should be relaxed. When using the rudder pedals, pressure should be applied smoothly and evenly by pressing with the ball of one foot. Since the rudder pedals are interconnected through springs or a direct mechanical linkage and act in opposite directions, when pressure is applied to one rudder pedal, foot pressure on the opposite rudder pedal should be relaxed proportionately.

In summary, during flight, the pressure the pilot exerts on the aileron and elevator controls and rudder pedals causes the airplane to move about the roll (longitudinal), pitch (lateral), and yaw (vertical) axes. When a control surface moves out of its streamlined position (even slightly), moving air exerts a force against that surface. It is this force that the pilot feels on the controls.

Feel of the Airplane

The ability to sense a flight condition, such as straight-and-level flight or a dive, without relying on instrumentation is often called “feeling the airplane.” Examples of this “feel” may be sounds of the airflow across the airframe, vibrations felt through the controls, engine and propeller sounds and vibrations at various flight attitudes, and the sensations felt by the pilot through physical accelerations.

Humans sense “feel” through kinesthesia (the ability to sense movement through the body) and proprioception (unconscious perception of movement and spatial orientation). These stimuli are detected by nerves and by the semicircular canals of the inner ear. When properly developed, kinesthesia can provide the pilot with critical information about changes in the airplane's direction and speed; however, there are limits in kinesthetic sense when relied upon solely without visual information, as when flying in instrument meteorological conditions (IMC). Sole reliance on the kinesthetic sense ultimately leads to disorientation and loss of aircraft control.

Developing this “feel” takes time and exposure in a particular airplane. It only comes with dedicated practice at the various flight conditions so that a pilot's senses are trained by the sounds, vibrations, and forces produced by the airplane. The following are some important examples:

- Rushing air creates a distinctive noise pattern and as the level of sound increases, it likely indicates that the airplane's airspeed is increasing and that the pitch attitude is decreasing. As the noise decreases, the airplane's pitch attitude is likely increasing and its airspeed decreasing.
- The sound of the engine in cruise flight is different from that in a climb and different again when in a dive. In fixed-pitch propeller airplanes, when the airplane's pitch attitude increases, the engine sound decreases and as pitch attitude decreases, the engine sound increases.
- In a banked turn, the pilot is forced downward into the seat due to the resultant load factor. The increased G force of a turn feels the same as the pull up from a dive, and the decreased G force from leveling out feels the same as lowering the nose out of a climb.

Sources of actual “feel” are very important to the pilot. This actual feel is the result of acceleration, which is simply how fast velocity is changing. Acceleration describes the rate of change in both the magnitude and the direction of velocity. These accelerations impart forces on the airplane and its occupants during flight. The pilot can sense vertical forces through pressure changes into the seat or horizontal forces while being pushed from side to side in the seat if the airplane slips or skids. These forces need not be strong, only perceptible by the pilot, to be useful. An accomplished pilot who has excellent “feel” for the airplane is able to detect even the smallest accelerations.

The flight instructor should teach the difference between perceiving and reacting to sound, vibrations, and forces versus merely noticing them. It is this increased understanding that contributes to developing a “feel” for the airplane. A pilot who develops a “feel” for the airplane early in flight training is likely to have less difficulty during more advanced training.

Attitude Flying

An airplane's attitude is determined by the angular difference between a specific axis and the natural horizon. A false horizon can occur when the natural horizon is obscured or not readily apparent. This is an important concept because it requires the pilot to develop a pictorial sense of this natural horizon. Pitch attitude is the angle formed between the airplane's longitudinal axis, which extends from the nose to the tail of the airplane, and the natural horizon. Bank attitude is the angle formed by the airplane's lateral axis, which extends from wingtip to wingtip, and the natural horizon. [Figures 3-3A and 3-3B] Angular difference about the airplane's vertical axis (yaw) is an attitude relative to the airplane's direction of flight but not relative to the natural horizon.

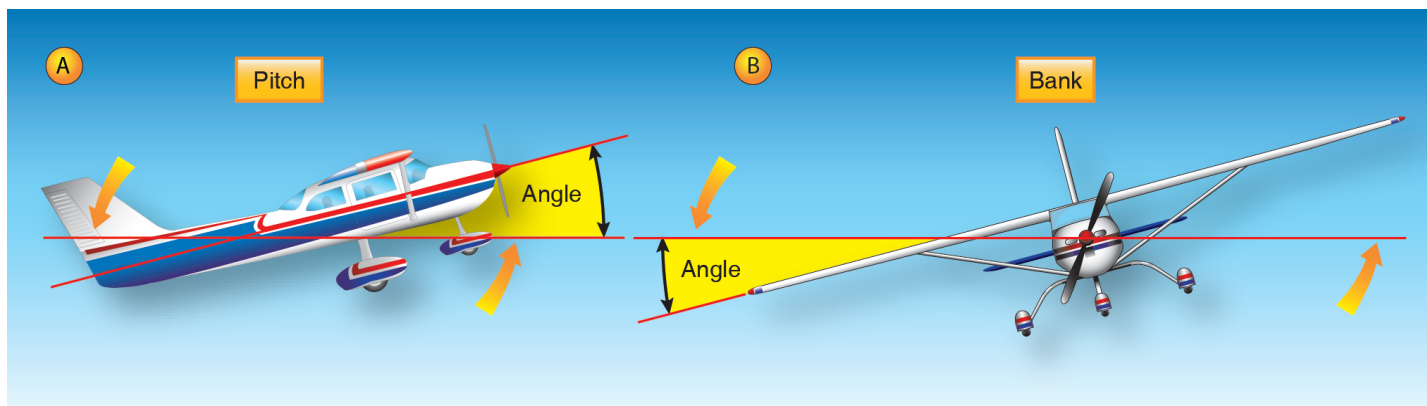


Figure 3-3. (A) Pitch attitude is the angle formed between the airplane's longitudinal axis, which extends from the nose to tail of the airplane, and the natural horizon. (B) Bank attitude is the angle formed by the airplane's lateral axis, which extends from wingtip to wingtip, and the natural horizon.

Controlling an airplane requires one of two methods to determine the airplane's attitude in reference to the horizon. When flying "visually" in visual meteorological conditions (VMC), a pilot uses their eyes and visually references the airplane's wings and cowling to establish the airplane's attitude to the natural horizon (a visible horizon). If no visible horizon can be seen due to clouds, whiteouts, haze over the ocean, night over a dark ocean, etc., it is IMC for practical and safety purposes. [Figure 3-4] When flying in IMC or when cross-checking the visual references, the airplane's attitude is controlled by the pilot referencing the airplane's mechanical or electronically-generated instruments to determine the airplane's attitude in relation to the natural horizon.

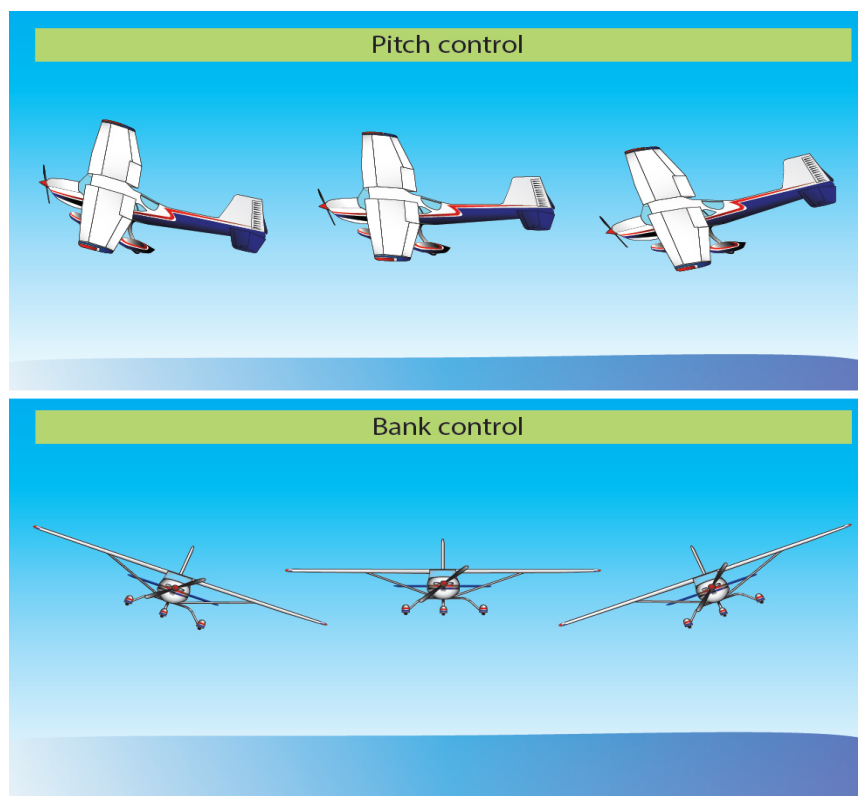


Figure 3-4. Airplane attitude is based on relative positions of the nose and wings on the natural horizon.